

SUBDOMAIN ENUMERATION TOOL – PROJECT SUMMARY

Overview

This project is a Python-based Subdomain Enumeration Tool developed to discover publicly accessible subdomains associated with a target domain.

The tool performs DNS lookups against a predefined list of common subdomains and identifies those that resolve successfully. This helps security professionals gain visibility into an organization's external attack surface and identify exposed services.

Features Added

- Automated subdomain discovery
- DNS-based hostname resolution
- Detection of active subdomains
- Professional ASCII banner interface
- User-friendly terminal output
- Result summary reporting
- Exception handling for DNS failures
- Lightweight and fast execution
- Attack surface reconnaissance support

Information Collected

- Discovered Subdomain Names
- Resolved IP Addresses
- Total Active Subdomains
- DNS Resolution Results

Technologies Used

- Python
- Socket Module
- DNS Resolution

Example Output

```
(root@kali)-[/home/kali/Desktop]
# python ssl_analyzer.py

OCTOBER
SSL Certificate Checker
Enter Domain (e.g. google.com): amazon.com

SSL CERTIFICATE INFORMATION
Domain      : amazon.com
Issuer      : DigiCert Inc
Valid From   : May  5 00:00:00 2026 GMT
Valid Until  : Nov 18 23:59:59 2026 GMT
Days Remaining : 162
Status      : Valid
```

Concepts Demonstrated

- Subdomain Enumeration
- DNS Fundamentals
- Attack Surface Discovery
- Information Gathering
- Network Reconnaissance
- Python Automation
- Cybersecurity Assessment Techniques

Learning Outcome

This project helps understand how cybersecurity professionals perform reconnaissance and identify publicly exposed assets before conducting security assessments.

Practical skills gained include:

- DNS Enumeration
- Asset Discovery
- External Attack Surface Mapping
- Host Identification
- Security Reconnaissance
- Python-Based Security Automation

Security Applications

- External Asset Discovery
- Attack Surface Assessment
- Security Auditing
- Reconnaissance Activities
- Infrastructure Visibility
- Vulnerability Assessment Preparation

Future Enhancements

- Wordlist-Based Enumeration
- Multi-threaded Scanning
- Wildcard DNS Detection
- DNS Record Enumeration
- Export Results to CSV/PDF
- WHOIS Integration
- Certificate Transparency Log Analysis
- Search Engine Reconnaissance

Disclaimer

This project is intended for educational purposes and authorized security assessments only. Always obtain proper permission before performing reconnaissance or security testing activities.

30 Days of Cybersecurity Automation – Day 08

Building practical cybersecurity tools through hands-on automation and real-world security assessments.

Author: Zeeshan Alam